

YOLO-6D-Pose: Enhancing YOLO for Single-Stage Monocular Multi-Object 6D Pose Estimation

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Abstract

Directly regressing 6 degrees of freedom for all the objects from a single RGB image is not well explored. Even end-to-end pose estimation approaches for a single object are inferior compared to state-of-the-art methods in terms of accuracy. Most 6D pose estimation frameworks are multi-stage relying on off-the-shelf deep networks for object and keypoint detection to establish correspondences between 3D object keypoints and 2D image locations. This is followed by applying a variant of a RANSAC-based Perspective-n-Point (PnP) followed by complex refinement operation. In this work, we propose a multi-object 6D pose estimation framework by enhancing the popular YOLOX object detector. The network is end-to-end trainable and detects each object along with its pose from a single RGB image without any additional post-processing. We show that by properly parameterizing the 6D pose and carefully designing the loss function, we can achieve state-of-the-art accuracy without further refinement or any intermediate representations. YOLO-6D-Pose achieves SOTA results on YCBV and LMO dataset, surpassing all existing monocular approaches. We systematically analyze various 6D augmentations to verify their correctness and propose a new translation augmentation for this task. The network does not rely on any correspondences and is independent of the CAD model during inference. Code is available at <https://github.com/TexasInstruments/edgeai-yolox>.

1. Introduction

Object pose estimation aims to estimate the 3D orientation and 3D translation of objects in a given environment. It is useful in a wide range of applications like robotic manipulation [2][28] for bin-picking, motion planning, and human-robot interaction task such as learning from demon-

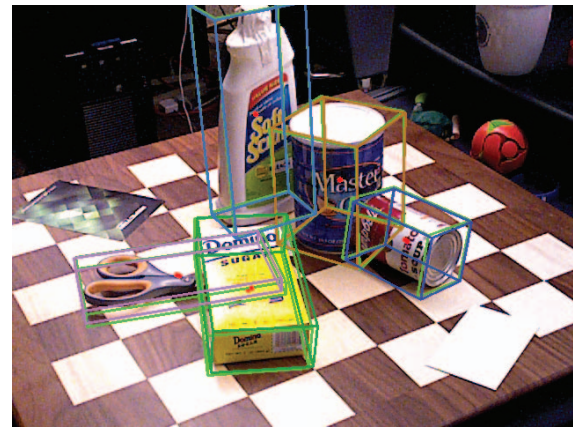


Figure 1. Example prediction of YOLOX-l-object-pose model performing end-to-end accurate 6d pose estimation for all the object in an image in a single forward pass in constant runtime. Green 3D bounding boxes visualize ground-truth poses while inferred poses are visualized with class specific colors.

stration. This problem is quite challenging due to factors like heavy occlusion, object symmetry, lighting variation, and a variety of objects with different shapes. Advances in deep learning have led to significant breakthrough in this problem. Most best-performing work relies on a multi-stage strategy of first establishing 2D-3D correspondences and then computing the 6D pose with a RANSAC based Perspective-n-Point(PnP) [7] algorithm [16][11][21]. Some approaches [23] use pixel-wise RANSAC based voting scheme to make keypoint detection robust. These methods cannot be trained end-to-end and may require extra computation for further refinement of the pose. Runtime for these approaches increase linearly with the number of objects in the image as we need to compute 6D pose via PnP for each object. Some methods [16][14] require an object detector to localize the object of interest that is fed to the 6D pose estimation engine which means that the entire chain needs

to be executed for each object separately.

In our work, we propose an approach to get rid of the above drawbacks and still achieve competitive accuracy. We have extended the popular YOLOX [8] family of object detection architecture in a natural way to predict the pose of each detected object. In a single forward pass of the network, we can detect each object and estimate its pose. Here, we regress a parameterized version of the 6D pose from which we can estimate the rotation and translation parameters with a simple transformation. There is no need for any further post-processing or complex refinement. The proposed method is compared with the participants of BOP challenge 2019 as well as other methods.

To summarize, we make the following contributions.

- We propose an end-to-end approach based on YOLOX that takes a single image as an input and infers accurate pose of each object without any intermediate representation or complex post-processing like refinement.
- We systematically analyze various 6D augmentations and mathematically verify their correctness. This allows us to correctly tune the augmentation parameters and propose a new translation augmentation.
- We improve the baseline ADD(-S) loss function by gradually incorporating various objectives to supervise individual pose components along with the holistic ADD(-S) metric.
- The method achieves SOTA result amongst all monocular RGB based approaches including refinement based approaches like [25] [17] and [14].

2. Related Works

Monocular 6D pose estimation can be partitioned into two distinct paradigms. Some methods try to directly regress the final 6D pose while others estimate pose from 2D-3D correspondences with RANSAC/PnP paradigm. Both approaches can use various refinement techniques to improve the initially estimated pose.

2.1. Indirect Methods

This is the most popular approach that relies on establishing 2D-3D correspondences before solving for the 6D pose using RANSAC based PnP algorithm. [24] and [27] compute the 2D projection of the 3D bounding box corners. Peng et al. [23] showed that keypoints away from the object surface result in larger errors and proposed to sample several keypoints on the object model. To enhance the robustness, [23] and [12] uses segmentation with voting for each correspondence. These are among the best performing approaches on several challenging benchmark.

2.2. Direct Methods

These methods directly regress a representation of the 6D pose. Kehl et al. [13] extends SSD [19] to estimate the 6D object pose by discretizing the pose space and conducting classification rather than regression. PoseCNN [31] regresses depth, the projected 2D center and a quaternion for each region-of-interest in a custom detection pipeline. It uses a Hough voting layer to localize the object center. DeepIM [15] proposed iterative refinement by regressing a pose difference between a render of the pose hypothesis and the input image. CosyPose [14] enhanced DeepIM by including a continuous rotation parameterization and using more recent architectures. EfficientPose [4] proposed an enhancement of EfficientDet [26] object detection model to perform pose estimation. However, due to the lack of parameterization of the pose parameters, use of vanilla ADD(-S) loss function and use of very strong scale-augmentation results in instability while training on large datasets like YCB-V [31].

3. Method

Given an RGB image I and a set of N objects $\mathcal{O} = \{\mathcal{O}_i | i = 1, \dots, N\}$ together with their corresponding 3D CAD models $\mathcal{M} = \{\mathcal{M}_i | i = 1, \dots, N\}$, our goal is to estimate the 6D pose $\mathbf{P} = [\mathbf{R} | \mathbf{t}]$ w.r.t the camera for each object $\mathcal{O}$ present in I . Note that $\mathbf{R}$ describes the 3D rotation and $\mathbf{t}$ denotes the 3D translation of the detected object.

Figure 2 represents a schematic overview of the proposed methodology. In the core, we enhanced YOLOX [8] object detector to perform 6D pose estimation. Unlike most other approaches that first detect an object and zoom in that ROI and feed it to other network to predict pose for that object via some intermediate representation or correspondences, our approach is truly end-to-end. The network is trained to directly predict the 2D bounding box and the corresponding pose for each object without relying on any intermediate representation or correspondences.

In the following, we first look into our choice of parameterization of the pose to make the learning process easier. This is followed by analyzing various 6D augmentations. Direct augmentation of 6D pose is difficult and constrained. E.g. certain transformations in the image domain may not result in any valid pose for the objects. We analyze in detail various augmentation techniques that were proposed by [4] to tune the augmentation parameters. We propose a new translation augmentation and study its impact on overall accuracy. We then look into our proposed loss function that contains a loss term for each pose parameter as well as a holistic ADD(-s) loss that couples the rotation and translation together.

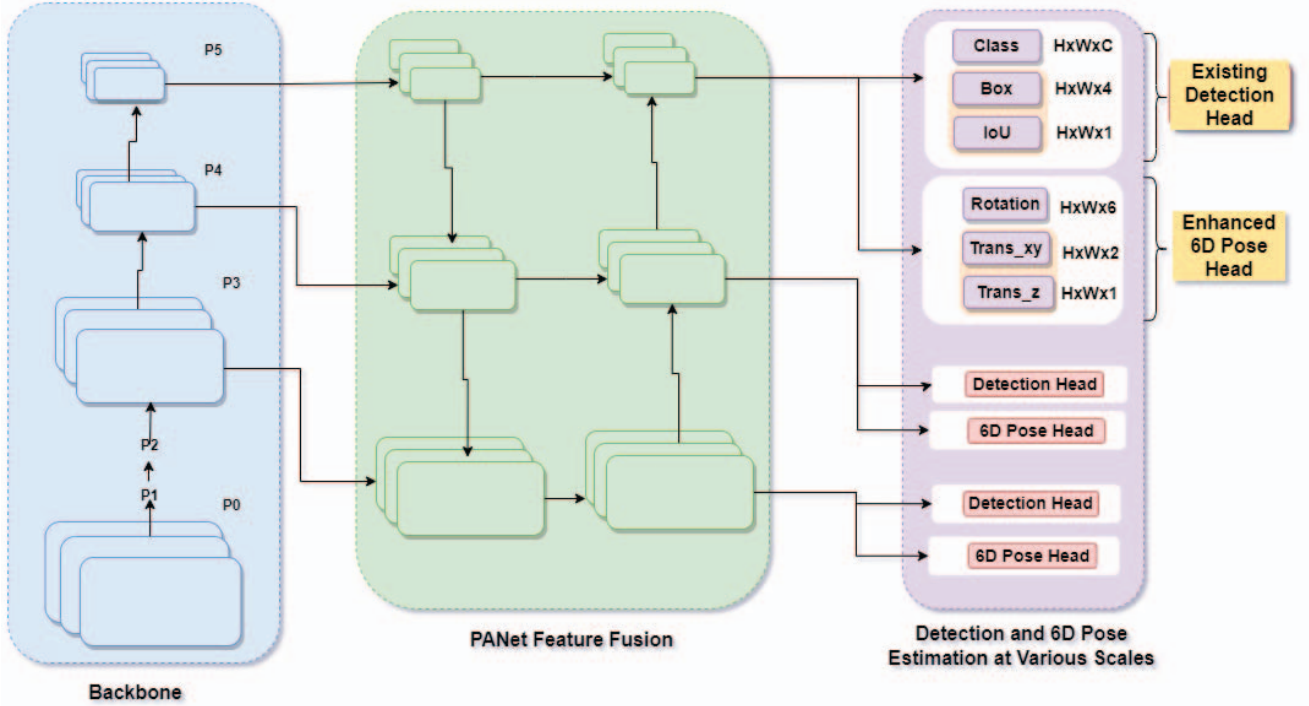


Figure 2. YOLO-pose architecture based on YOLOv5. Input image is passed through darknet53 backbone that generates feature maps at various scales P3, P4, P5, P6. PANet is used for fusing these feature maps across multiple scales. Output of PANet is fed to the detection and pose head. Pose head consists of Rotation, Trans_{xy} and Trans_z heads. Different components of Detection and Pose head are shown for the first head.

3.1. 6D Object Pose Parameterization

Single-stage and multi-stage direct 6D pose estimation methods usually differ in the parameterization of the rotation $\mathbf{R}$ and translation $\mathbf{t}$ parameters. In this section, we investigate different representations of the pose and finally select the one that can be easily integrated into a detection framework and lead to stable training.

Parameterization of 3D Rotation. Several different parameterizations can be employed to describe 3D rotations. Since many representations exhibit ambiguities, i.e. R_i and R_j describes the same rotation with $R_i \neq R_j$, most work relies on parameterizations that are unique to help training. Therefore, common choices are unit quaternions [31] [15] and log quaternions [22]. However, these representations for 3D rotation have discontinuities in Euclidean space. While regressing, this introduces an error close to the discontinuities. To overcome this limitation, [32] proposed a novel continuous 6-dimensional representation for $\mathbf{R}$ in $SO(3)$, which has been proven promising [14]. The 6-dimensional representation $\mathbf{R}_{6d}$ is defined as the first two columns of $\mathbf{R}$.

$$\mathbf{R}_{6d} = [\mathbf{R}_{.1}, \mathbf{R}_{.2}] \quad (1)$$

Given a 6-dimensional predicted vector $R_{6d} = [r_1, r_2]$, the rotation matrix $\mathbf{R} = [\mathbf{R}_{.1} | \mathbf{R}_{.2} | \mathbf{R}_{.3}]$ can be computed according to

$$\begin{aligned} \mathbf{R}_{.1} &= \phi(\mathbf{r}_1) \\ \mathbf{R}_{.2} &= \phi(\mathbf{r}_2) - (\mathbf{R}_{.1} \cdot \phi(\mathbf{r}_2))\mathbf{R}_{.1} \\ \mathbf{R}_{.3} &= \mathbf{R}_{.1} \times \mathbf{R}_{.2} \end{aligned} \quad (2)$$

Where $\phi(\cdot)$ is the normalization operation. Given the advantage of this representation, we employ $\mathbf{R}_{6d}$ to parameterize the 3D rotation.

Parameterization of 3D Translation. Instead of directly regressing the translation $\mathbf{t} = [t_x, t_y, t_z]^T \in \mathbb{R}^3$ in 3D space, previous works decoupled the translation into the 2D location (c_x, c_y) of the projected 3D centroid and the object's distance t_z towards the camera. Given the camera intrinsics $\mathbf{K}$, the 3D translation can be recomputed as:

$$\mathbf{t} = \mathbf{K}^{-1} \mathbf{t}_z [c_x, c_y, 1]^T \quad (3)$$

[13] approximate (c_x, c_y) as the bounding box center and estimate t_z using a reference camera distance. [31] directly regresses the (c_x, c_y) and t_z . To take advantage of the object detection framework, We make this representation scale and

location invariant. In YOLOX, the model predicts at multiple scales w.r.t to some specified grid locations. If an object is matched against a particular grid location (g_x, g_y) at stride of (s_x, s_y) , the translation parameters are transformed as:

$$\begin{aligned}\delta_x &= (c_x/s_x - g_x) \\ \delta_y &= (c_y/s_y - g_y) \\ \delta_z &= \log(t_z)\end{aligned}\quad (4)$$

t_z is normalized log-normalized to reduce its range. Finally, the 3D translation can be recovered from $[\delta_x, \delta_y, \delta_z]^T$ by simply following the inverse transformation of Equation 3 and Equation 4.

3.2. YOLOX based Direct Regression Network

In this section, we present our YOLOX [8] based architecture which is referred as YOLOX-6D-Pose. Unlike the models that first detect each object followed by estimating the 6D pose for each object, YOLOX-6D-Pose learns to perform object detection and pose estimation simultaneously, making our approach truly end-to-end.

Network Architecture. YOLOX is a preferred choice for our base detector as it provides a strong performance-accuracy trade-off for object detection. Our goal is to enhance the architecture with minimal additional cost of computation. The entire network is based on CSPDarknet53 [29] backbone and PANet [18] based feature aggregation. Corresponding to each detection head of the network we introduce a rotation and translation head as shown in Figure 2.

Rotation Head. Rotation head is trained to predict the 6d representation of the rotation parameters as described in Equation 1. This is followed by performing the post-processing operations as described by Equation 2 to decode the final rotation matrix. Even though our approach can directly learn to predict the rotation vector $r \in \mathbb{R}^3$ as in [4], this approach is robust and results in significant stability during training. Again, unlike approaches that rely on an iterative refinement module, our rotation head does not rely on any such technique.

Translation Head. The Translation head is subdivided into T_{xy} and T_z heads. T_{xy} head predicts $[\delta_x, \delta_y]$ whereas T_z head predicts $[\delta_z]$. Final translation parameters are recovered following the inverse of Equation 3 and Equation 4.

Unifying Detection and 6D Pose Estimation. Unlike many other two-stage or multi-stage models that try to decouple the task of detection and 6D pose estimation, this work unifies these two tasks. During training, any anchor or grid that matches with the ground-truth box stores the entire pose of that object. During inference, we predict the pose for each detected object in a single forward pass. Final detections are filtered out via non-max suppression of overlapping boxes.

3.3. 6D Augmentation

End-to-end 6D augmentation deals with augmenting an image and directly transforming the 6D Pose of all objects present in that image. This is complicated than augmenting keypoints in 2D space that is widely used in most two-stage approaches. It was proposed by [4] to tackle the limited data of LM-O dataset [3]. They proposed rotation and scale augmentation to improve accuracy on LM-O. We will look into these two augmentations along with the proposed translation augmentation.

Mathematically, an augmentation is correct if for a transformation in 3D space of any points $x \in M$ in an object CAD model, we can find a corresponding transformation in the image space such that all the points of the 3D CAD model can be projected on the image without any loss of generality. Figure 3 shows a visualization of CAD models projected on an image with the ground-truth pose. We will verify all 6D augmentations with this visualization.



Figure 3. Projection of 3D CAD models on an image using ground-truth pose without any augmentation

Rotation Augmentation. In rotation augmentation, we rotate the image along the principal axis and all the objects pose are transformed by the same rotation around the z axis. The rotation around the z axis can be described with the rotation vector $\Delta\theta$ in axis angle representation as follows:

$$\Delta\theta = [0, 0, \theta] \quad (5)$$

E.g. Consider an object with ground-truth pose $[R|t]$ and a sample point on the object w.r.t object's frame of reference is $X_o = [x_o, y_o, z_o]$. This point can be represented w.r.t the camera's frame of reference X_c as:

$$X_c = RX_o + t \quad (6)$$

Let ΔR be the rotation matrix corresponding to $\Delta\theta$. Upon augmentation, this point becomes:

$$\Delta R X_c = \Delta R R X_o + \Delta R t \quad (7)$$

Hence, the new augmented $[\mathbf{R}_{\text{aug}}|\mathbf{t}_{\text{aug}}]$ becomes:

$$\begin{aligned}\mathbf{R}_{\text{aug}} &= \Delta\mathbf{r}\mathbf{R} \\ \mathbf{t}_{\text{aug}} &= \Delta\mathbf{r}\mathbf{t}\end{aligned}\quad (8)$$

Now, let's find out the augmentation parameters in homogeneous coordinates where we can write

$$\begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix} = \begin{bmatrix} x_{hc} \\ y_{hc} \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11}x_o + r_{12}y_o + r_{13}z_o + t_x \\ r_{31}x_o + r_{32}y_o + r_{33}z_o + t_z \\ r_{21}x_o + r_{22}y_o + r_{23}z_o + t_y \\ r_{31}x_o + r_{32}y_o + r_{33}z_o + t_z \\ 1 \end{bmatrix}\quad (9)$$

With a generic rotation transformation in 3D space, we get:

$$\begin{bmatrix} x'_c \\ y'_c \\ z'_c \end{bmatrix} = \begin{bmatrix} \Delta r_{11} & \Delta r_{12} & \Delta r_{13} \\ \Delta r_{21} & \Delta r_{22} & \Delta r_{23} \\ \Delta r_{31} & \Delta r_{32} & \Delta r_{33} \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix}\quad (10)$$

In homogeneous coordinate it becomes:

$$\begin{bmatrix} x'_c \\ y'_c \\ z'_c \\ 1 \end{bmatrix} = \begin{bmatrix} \Delta r_{11}x_c + \Delta r_{12}y_c + \Delta r_{13}z_c \\ \Delta r_{21}x_c + \Delta r_{22}y_c + \Delta r_{23}z_c \\ \Delta r_{31}x_c + \Delta r_{32}y_c + \Delta r_{33}z_c \\ 1 \end{bmatrix} \neq \Delta\mathbf{r} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix}\quad (11)$$

As clear from the above equation, we cannot simply transform the points in homogeneous coordinate by $\Delta\mathbf{r}$ to get the corresponding transformed points. However, with the constraint of rotation around the z axis, we have

$$\begin{bmatrix} x'_c \\ y'_c \\ z'_c \end{bmatrix} = \begin{bmatrix} \Delta r_{11} & \Delta r_{12} & 0 \\ \Delta r_{21} & \Delta r_{22} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \begin{bmatrix} \Delta r_{11}x_c + \Delta r_{12}y_c \\ \Delta r_{21}x_c + \Delta r_{22}y_c \\ z_c \end{bmatrix}\quad (12)$$

$$\begin{bmatrix} x'_c \\ y'_c \\ z'_c \\ 1 \end{bmatrix} = \begin{bmatrix} \Delta r_{11} \frac{x_c}{z_c} + \Delta r_{12} \frac{y_c}{z_c} \\ \Delta r_{21} \frac{x_c}{z_c} + \Delta r_{22} \frac{y_c}{z_c} \\ 1 \end{bmatrix} = \Delta\mathbf{r} \begin{bmatrix} \frac{x_c}{z_c} \\ \frac{y_c}{z_c} \\ 1 \end{bmatrix}\quad (13)$$

Hence, both the ground-truth pose and the 2D image are transformed by the same rotation matrix $\Delta\mathbf{r}$. This augmentation is valid only for rotation around the z axis. For rotation around X or Y axis, we cannot find any 6D pose parameter to define the augmentation in the homogeneous domain. This augmentation is mathematically correct and the projected CAD models perfectly align with the objects in augmented image as shown in **Figure 4**.

Scale Augmentation. Scale augmentation resizes the image by a factor s and scales the t_z component of the pose by the same scale to $s \cdot t_z$. However, it is mathematically incorrect for various reasons:

- **Occlusion:** An object that is visible at a depth t_z may not be visible at a depth of $t_z \cdot s$.



Figure 4. Image rotated by 20 deg around the principal axis.

- **Objects with High Rotation:** Scale augmentation can be incorrect even in absence of occlusion. The discrepancy in scale augmentation is negligible if the rotation component of the pose is small. However, it becomes prominent when the rotation is significantly high.

With scale augmentation s , as proposed in [4], the points in homogeneous coordinate becomes:

$$\begin{bmatrix} x'_c \\ y'_c \\ z'_c \\ 1 \end{bmatrix} = \begin{bmatrix} x'_{hc} \\ y'_{hc} \end{bmatrix} = \begin{bmatrix} r_{11}x_o + r_{12}y_o + r_{13}z_o + \frac{t_x}{s} \\ r_{31}x_o + r_{32}y_o + r_{33}z_o + \frac{t_z}{s} \\ r_{21}x_o + r_{22}y_o + r_{23}z_o + \frac{t_y}{s} \\ r_{31}x_o + r_{32}y_o + r_{33}z_o + \frac{t_z}{s} \end{bmatrix}\quad (14)$$

From **Equation 9** and **Equation 14** it is clear that:

$$\begin{bmatrix} x_{hc} \\ y_{hc} \end{bmatrix} \neq s \cdot \begin{bmatrix} x'_{hc} \\ y'_{hc} \end{bmatrix}\quad (15)$$

Hence, just scaling t_z to $\frac{t_z}{s}$ does not result in correct augmentation. It is not possible to parameterize scale augmentation by scaling the t_z component of the pose. If the object rotation is significantly small, the pose can be approximated just by scaling the t_z component. In this case, the conditions below are satisfied:

$$\begin{aligned}r_{31}x_o + r_{32}y_o + r_{33}z_o &\ll t_z/s \\ r_{11}x_o + r_{12}y_o + r_{13}z_o &\ll t_x \\ r_{21}x_o + r_{22}y_o + r_{23}z_o &\ll t_y\end{aligned}\quad (16)$$

If all of the above conditions are satisfied, we have

$$\begin{bmatrix} x_{hc} \\ y_{hc} \end{bmatrix} = s \cdot \begin{bmatrix} x'_{hc} \\ y'_{hc} \end{bmatrix}\quad (17)$$

One important point to note is that the first condition of **Equation 16** is satisfied even with large rotation since t_z is in general a much larger quantity than t_x and t_y . However, the other two conditions are satisfied only with small rotation. We observe a significant drop with scale augmentation uniformly sampled from $[0.7, 1.3]$ as proposed in [4].



Figure 5. Image scaled by a factor of 1.3 across both height and width. We can see a misalignment of the objects with their projected CAD models.

Figure 5 is a visualization of scale augmentation.

Translation Augmentation. Keeping the constraints of Equation 16 in mind, we introduce this new augmentation in our training where we randomly translate the image horizontally and vertically and accordingly adjust the t_x and t_y component of the pose. For translation of a pixel by $[\Delta_x, \Delta_y]$ in the homogeneous domain, the proposed augmentation augment the t_x and t_y component of the pose for an object by $[t_z \Delta_x, t_z \Delta_y]$.

$$\begin{aligned} \begin{bmatrix} x'_{hc} \\ y'_{hc} \end{bmatrix} &= \begin{bmatrix} \frac{r_{11}x_o + r_{12}y_o + r_{13}z_o + t_x + \Delta_x t_z}{r_{31}x_o + r_{32}y_o + r_{33}z_o + t_z} \\ \frac{r_{21}x_o + r_{22}y_o + r_{23}z_o + t_y + \Delta_y t_z}{r_{31}x_o + r_{32}y_o + r_{33}z_o + t_z} \end{bmatrix} \\ &= \begin{bmatrix} x_{hc} \\ y_{hc} \end{bmatrix} + \begin{bmatrix} \frac{\Delta_x t_z}{r_{31}x_o + r_{32}y_o + r_{33}z_o + t_z} \\ \frac{\Delta_y t_z}{r_{31}x_o + r_{32}y_o + r_{33}z_o + t_z} \end{bmatrix} \end{aligned} \quad (18)$$

As discussed earlier, in most of the cases, the first condition of Equation 16 holds. Hence, we have

$$\begin{bmatrix} x'_{hc} \\ y'_{hc} \end{bmatrix} = \begin{bmatrix} x_{hc} + \Delta_x \\ y_{hc} + \Delta_y \end{bmatrix} \quad (19)$$

Figure 6 is a visualization of translation augmentation.

Augmentation Parameters. Final augmentation is composed of a random uniform selection of rotation from (0, 10) degrees, translation of (0, 10) %, and scaling factor from (0.9, 1.1). We reduce the scale augmentation based on our observation in Figure 5. Random color-space augmentation is always used as it does not require any transformation of pose parameters. Figure 7 is a visualization of combining all the augmentations.

3.4. Loss Function

The choice of the loss function is crucial for 6D pose estimation. Most work [31] employ a variant of Point-Matching



Figure 6. Image translated by 10 % across height and width. The projected CAD models align reasonably well with the objects



Figure 7. Augmented image with 10 degree rotation, 10 % translation, and a scale factor of 1.1.

Loss based on the ADD(-S) metric to couple rotation and translation. We use a combination of various loss functions to optimize the pose parameters as a whole with ADD(-S) as well as optimize each component of the pose independently to overcome the drawbacks of ADD(-S) loss.

ADD(-S) Loss. Instead of considering only the rotation as in PoseLoss [31] [30], we try to optimize the ADD(-S) metric as proposed in [15][4]. $\mathcal{L}_{ADD(-S)}$ is defined as:

$$\begin{cases} \mathcal{L}_{asym} = \frac{1}{m} \sum_{x \in m} \left\| \frac{(\mathbf{R}_p \mathbf{x} + \mathbf{t}_p) - (\mathbf{R}_g \mathbf{x} + \mathbf{t}_g)}{d_m} \right\|_2 \\ \mathcal{L}_{sym} = \frac{1}{m} \sum_{x_1 \in m} \min_{x_2 \in m} \left\| \frac{(\mathbf{R}_p \mathbf{x}_1 + \mathbf{t}_p) - (\mathbf{R}_g \mathbf{x}_2 + \mathbf{t}_g)}{d_m} \right\|_2 \end{cases} \quad (20)$$

$\mathbf{R}_g$ and $\mathbf{t}_g$ defines the ground-truth pose whereas $\mathbf{R}_p$ and $\mathbf{t}_p$ defines the predicted pose. M denotes the set of the object's 3D model points, m is the number of points and d_m is the object diameter. This approach has the advantage that the model is optimized directly on a metric itself. However,

ADD(-S) loss is dependent on the geometry of the object model and the sampling density of the surface. $\mathcal{L}_{sym}$ is similar to $\mathcal{L}_{asym}$, but instead of strictly calculating the distance between matching points of the two transformed point sets, the minimal distance for each point to any point in the transformed point set is taken into account. $\mathcal{L}_{sym}$ can yield unintuitively low errors even though the estimated poses are incorrect. Thus, $\mathcal{L}_{sym}$ loss is too lenient for symmetric objects.

Translation Loss. Since, our proposed network predicts the projection of $[t_x, t_y]$ on the image, we add a loss term to make the 2D prediction accurate. This task is similar to predicting a keypoint per object. Hence, we use a simplified version of Object Keypoint Similarity (OKS) loss which has been proven to be quite effective by [20]. $\mathcal{L}_{oks}$ is defined as:

$$\mathcal{L}_{OKS} = 1 - OKS = 1 - \exp\left(-\frac{d^2}{2s_b^2k^2}\right) \quad (21)$$

where, d is the Euclidean distance between ground-truth and the predicted projection of the 3D centroids. s_b is the area of the object and keypoint specific weight k is empirically set to 0.1. For the t_z component of the loss, we use Absolute relative difference (ARD) as an additional loss component in line with depth learning frameworks. ARD loss is defined as:

$$\mathcal{L}_{ARD} = 1 - \frac{t_{zp}}{t_{zg}} \quad (22)$$

where, t_{zp} and t_{zg} are the prediction and ground-truth for t_z . Finally, the overall pose loss with empirically chosen loss-specific weights $\lambda_{ADD(-S)}$, λ_{rot} , λ_{OKS} and λ_{ARD} becomes:

$$\mathcal{L}_{pose} = \lambda_{ADD(-S)}\mathcal{L}_{ADD(-S)} + \lambda_{rot}\mathcal{L}_{rot} + \lambda_{OKS}\mathcal{L}_{OKS} + \lambda_{ARD}\mathcal{L}_{ARD} \quad (23)$$

4. Experiments

In this section, we first introduce our experimental setup and then present the evaluation results for YCB-Video [31] and LINEMOD-Occluded [3] benchmark datasets.

Metrics. We evaluate our method based on the BOP challenge [10] and ADD(-S) 0.1d metrics. BOP evaluates three pose errors based on Visible Surface Discrepancy(VSD), maximum symmetry-aware surface distance(MSSD), and maximum symmetry-aware projection distance (MSPD). An average recall is computed for each of the errors, AR_{VSD} , AR_{MSSD} , AR_{MSPD} , based on a set of error thresholds, and the average recall, (AR), refers to the average of the three average recalls. Note that all BOP based results are taken from **YCB-V** and **LM-O** leaderboard.

Implementation Details. We use SGD optimizer with a cosine scheduler. The base learning rate is set to 1e-2. We train the model for 300 epochs.

Method	AR	AR _{VSD}	AR _{MSSD}	AR _{MSPD}
YOLOX-x-6D	83.5	79.3	85.9	85.4
ZebraPose [25] †	83.0	77.3	84.4	87.3
CIR [17] †	82.4	78.3	83.5	85.2
CosyPose [14] †	82.1	77.2	84.2	85.0
YOLOX-l-6D	81.3	76.7	83.6	83.6
SC6D [5]	76.5	69.5	79.6	80.4
YOLOX-m-6D	76.3	71.1	77.0	80.8
SurfEmb [9] †	71.8	x	x	x
YOLOX-s-6D	70.8	65.9	72.3	74.2
EPOS [11]	69.6	62.6	67.7	78.3
CosyPose [14]	65.5	x	x	x
Pix2Pose [21]	45.7	37.2	42.9	57.1

Table 1. Comparison with state-of-the-art monocular methods on YCB-V under BOP metrics. Proposed models outperform even methods relying on rendering-based refinement (annotated with †).

Method	ADD(-S)	AUC of ADD(-S)	Runtime (mS)
YOLOX-x-6D	88.5	88.9	29.8
YOLOX-l-6D	83.7	86.7	25.1
ZebraPose [25]	80.5	85.3	250.0
CosyPose [14]	-	84.5	320.0
YOLOX-m-6D	72.8	85.1	21.2
YOLOX-s-6D	66.3	84.4	16.9
DeepIM [15]	-	81.9	-
GDR-Net [30]	60.1	84.4	35.0
SO-Pose [6]	56.8	83.9	50.0
T6D-Direct [1]	48.7	74.6	17.0

Table 2. Comparison based on ADD(-S) metric on YCB-V dataset. Inference time for each model is reported alongside.

4.1. Evaluation on YCB-V Dataset

For YCB-V dataset, our models are trained on real and pbr data [10]. We compare our methods with existing monocular approaches in terms of the BOP and ADD(-S)0.1d metrics. We outperform refinement based method such as CosyPose [14], ZebraPose [25] as shown in **Table 1** and **Table 2**.

ADD(-S) based evaluation as in **Table 2** is required to enable comparison against methods like GDR-Net [30], T6D-Direct [1] that haven't been evaluated on BOP benchmark.

4.2. Evaluation on LINEMOD-Occluded Dataset

The real images in the training split of the LM-O dataset [3] have multiple objects. However, only one of them is annotated. Multi-stage methods like CosyPose [14] first ex-

tract the object ROI and then find the pose for each object independently. Since it trains one network per object, missing annotation is not an issue. However, YOLO-6D-Pose tries to predict the pose of all objects in the image in a single forward pass. Hence, it requires each object to have ground-truth annotation. Similar to CosyPose, We can as well train one network per object. However, that becomes inefficient and thus not explored in our work. We train exclusively using the 50K PBR training images provided as part of the BOP benchmark and achieve competitive accuracy compared to approaches trained with similar dataset. Results are presented in Table 3.

Method	AR	AR _{VSD}	AR _{MSSD}	AR _{MSPD}
CosyPose[14]	63.3	48.0	60.6	81.2
YOLOX-x-6D	62.9	48.3	61.1	79.4
CDPN [16]	62.4	46.9	61.2	73.1
YOLOX-l-6D	61.5	46.3	58.5	79.7
YOLOX-m-6D	59.4	44.1	56.2	77.9
YOLOX-s-6D	56.1	40.2	51.2	77.0
EPOS [11]	54.7	38.9	50.1	75.0

Table 3. Comparison with SOTA monocular methods trained with PBR images on LM-O dataset

4.3. Ablation Study: 6D Augmentation

To show the impact of each augmentation, we gradually enable each of the augmentations and check for its impact on the final accuracy for YOLOX-s-6d model on YCB-V dataset. All augmentation parameters are set as described in subsection 3.3. We report the results in Table 4.

Rotation	Translation	Scaling	AR
N	N	N	66.3
Y	N	N	68.6
N	Y	N	67.6
N	N	Y	67.1
Y	Y	Y	70.8

Table 4. Impact of each augmentation on the final accuracy.

4.4. Ablation Study: Loss Function

We quantify the impact of each component of the loss function on the overall accuracy. Compared to the baseline ADD(S) loss, using our composite loss results in significant improvement in accuracy as shown in Table 5. Specifically, OKS loss has the most significant impact over the baseline ADD(-S) loss.

ADD(-S)	OKS	ARD	AR
Y	N	N	66.1
N	Y	Y	49.9
Y	Y	N	69.7
Y	Y	Y	70.8

Table 5. Impact of various loss components on the final accuracy

4.5. Comparison with EfficientPose

In this work [4], the authors used a different train and test split of the LM-O dataset as discussed in this thread. Hence, their LM-O results cannot be used as a reference. However, our ablation study with similar loss function and augmentation parameters as EfficientPose shows significant improvement for our method.

4.6. Model Efficiency

We tested the runtime on a desktop with an Intel 3.60GHz CPU and an Nvidia 2080Ti GPU. In 6d pose estimation, very few work report the overall computation of the entire pipeline as it is not straightforward to estimate the computation of the rendering process in refinement-based methods. Hence, We report the end-to-end runtime for various methods for batchsize=1 in Table 2. We list down the compute and params of our models in Table 6. As evident from Table 2, for comparable accuracy, our models are much faster. Note that for approaches requiring multiple forward passes, we have reported the runtime for a single pass.

Method	GFLOPS	Params(M)
YOLOX-s-6D	31.2	11.6
YOLOX-m-6D	80.3	31.3
YOLOX-l-6D	161.2	64.8
YOLOX-x-6D	281.0	115.6

Table 6. Compute and latency of our proposed models

5. Conclusion

We have presented an end-to-end YOLOX based joint detection and pose estimation framework. Our models outperform most SOTA approaches at significantly lower complexity. This work is a coherent step toward unifying the field of object detection and pose estimation. We have carefully looked at each aspect and made suitable changes to push a simple architecture to achieve SOTA results in monocular object pose estimation. No other single-stage method has achieved that yet.

We believe, our results will further encourage the community to explore the potential of jointly solving these two tasks.

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